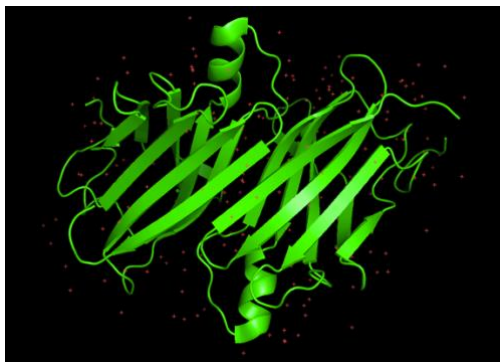


Validation and Correction of Protein Structural Models

Refinement of the crystal structure of Transthyretin (PDB: 1TTA)

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Part I: Structure Quality & Metadata

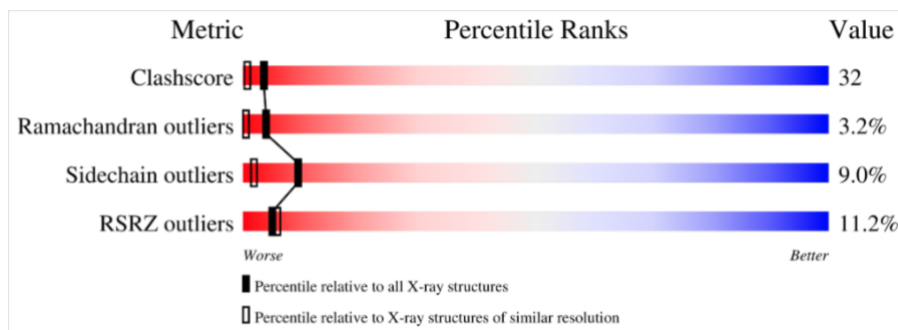


1. PDB Information

- **PDB ID:** 1TTA
- **Title:** THE X-RAY CRYSTAL STRUCTURE REFINEMENTS OF NORMAL HUMAN TRANSTHYRETIN AND THE AMYLOIDOGENIC VAL30MET VARIANT TO 1.7 ANGSTROMS RESOLUTION
- **Organism:** [Homo sapiens](#)
- **Year:** 1992
- **Study:** <https://doi.org/10.2210/pdb1TTA/pdb>

2. Methodological Quality

- **PDB ID:** 1TTA
- **Method:** X-RAY DIFFRACTION
- **Resolution:** 1.70 Å
- **R-values:**
 - R-Value Work: 0.170
 - R-Value Observed: 0.168



Transthyretin (1TTA) – a homotetrameric protein, which plays crucial role in amyloidosis. Despite high resolution in 1.70 Å, this protein shows poor metrics in compare to structures with similar resolution. All key indicators are in red zone – Worse – which indicates low geometric quality of the protein model.

Clashscore=32 is extremely high value, which means that structure is physically overcrowded and the atoms overlap. For high quality structures this value usually <10.

Ramachandran outliers=3.2% means that backbone of protein is broken or twisted to unnatural angles. A value <0.2% is normal.

Sidechain outliers=9.0% means that side chains are modelled incorrectly. Its can be result of automated modelling without verification.

RSRZ outliers=11.2% means that significant number of atoms located in regions with no electron density or shifted.

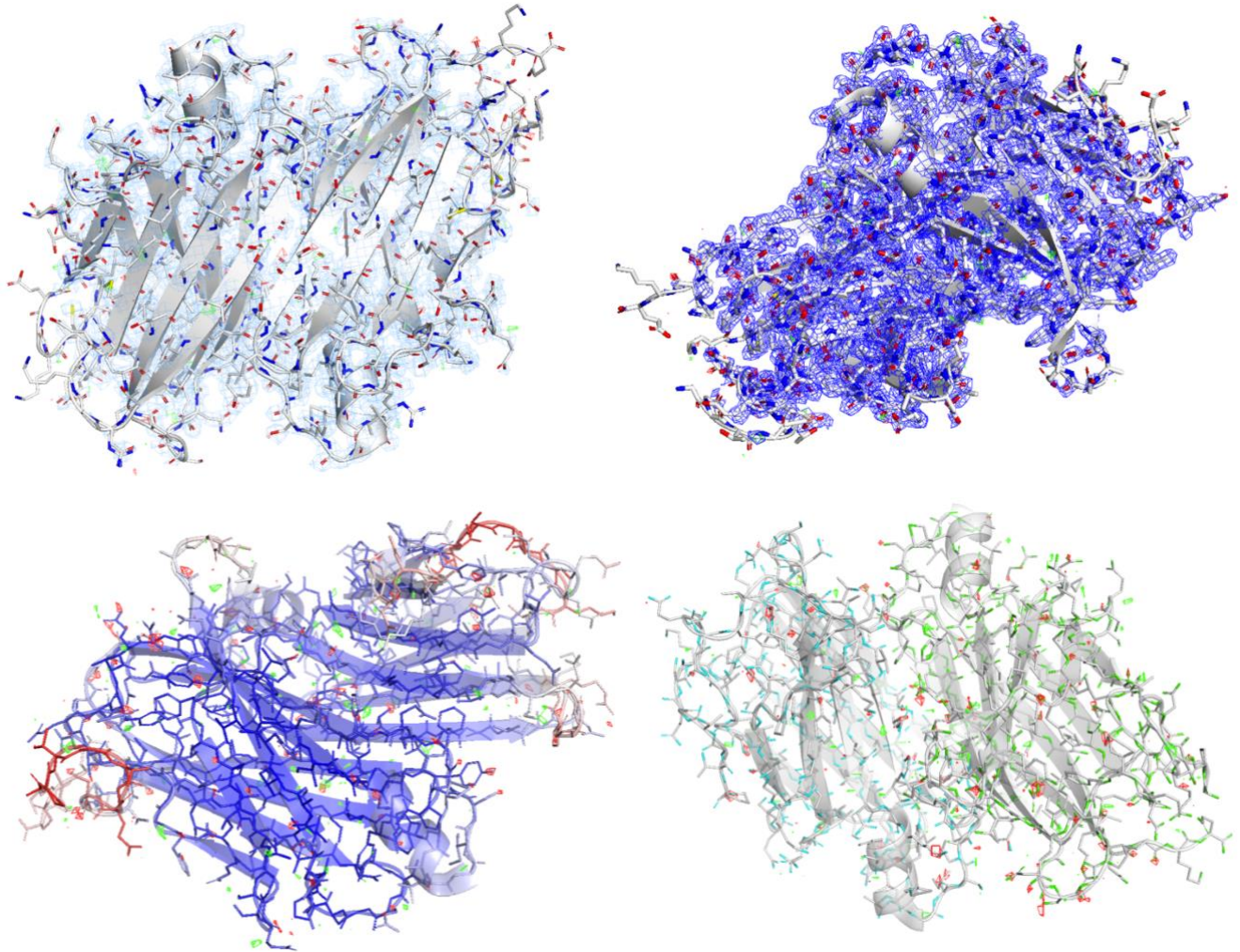
This values confirms that 1TTA structure requires refinancing to meet modern quality standart.

3. Visual analysis of density maps

Whtn overlying difference electron dencity maps Fo-Fe, a large amount of noise was observed:

- Red coluds (negative) indicates that atoms in the model were located not in right place, eg steric conflicts or false water molecules

- Green clouds (positive) indicates uncounted electron density where atoms absent



The core (rigid) part of tetramere has a clear density which allows to determine position of atoms clearly. However, peripheral (flexible) parts contain a lot of noise and density gaps which correlates with the main localization geometric errors (outliers).

Part II: Diagnostics using MolProbity

Diagnostics using MolProbity confirmed problematic protein geometry.

Clashscore: 16.29 (31th percentile) – for every 1000 atoms there are more than 16 serious clashes with resolution $>0.4\text{\AA}$.

Rotamer outliers: 9.65% (22 residues) -- ~one in ten side chains in energetically forbidden conformation

Critical contacts: physically impossible interaction $>1.0\text{\AA}$, eg side chain Thr3 and main chain Glu62 (1.07Å), or intrachain conflict in B-chain between Pro2 and Ser50 (1.096Å).

Also, a significant portion of conflicts (10.75%) belonged to protein-water interactions, which indicates that automated water picking didn't account v_n der Waals radii.

Moreover, torsion angles analysis showed 8 residues in Ramachandran map forbidden zone. So, protein backbone unnatural twisted. This errors often located in flexible loops or C/N-terminus with blurred electron density. So, low MolProbity score=2.92 places 1TTA structure to 4th percentile (96% of structures are better).

Almost 10% of all side chains are poor rotamers, eg B-chain has 4 outliers in a row – 80-86 residues region – Lys80, Leu82, Ser85, Pro86.

In addition, this structure has problems with bond lengths (1.13%) and angles (2.10%) – 23 covalent bonds deviate, and 58 angles have unacceptable values.

Summary statistics

All-Atom Contacts	Clashscore, all atoms:	16.29	31 st percentile* (N=819, 1.70Å ± 0.25Å)
	Clashscore is the number of serious steric overlaps (> 0.4 Å) per 1000 atoms.		
Protein Geometry	Poor rotamers	22	9.65% Goal: <0.3%
	Favored rotamers	183	80.26% Goal: >98%
	Ramachandran outliers	8	3.20% Goal: <0.05%
	Ramachandran favored	232	92.80% Goal: >98%
	Rama distribution Z-score	-0.54 ± 0.50	Goal: abs(Z score) < 2
	MolProbity score [^]	2.92	4 th percentile* (N=9248, 1.70Å ± 0.25Å)
	Cβ deviations >0.25Å	1	0.43% Goal: 0
	Bad bonds:	23 / 2028	1.13% Goal: 0%
Peptide Omegas	Bad angles:	58 / 2763	2.10% Goal: <0.1%
	Cis Prolines:	0 / 16	0.00% Expected: ≤1 per chain, or ≤5%
Additional validations	Chiral volume outliers	0/310	
	Waters with clashes	20/186	10.75% See UnDowser table for details

In the two column results, the left column gives the raw count, right column gives the percentage

#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	Bond lengths	Bond angles	Ci
			Avg: 24.33	Clashscore: 16.29	Outliers: 8 of 250	Poor rotamers: 22 of 228	Outliers: 1 of 234	Outliers: 16 of 272	Outliers: 39 of 272	Non-1
A 1		GLY	50.31	-	-	-	-	-	-	
A 2		PRO	50.29	-	OUTLIER (0.02%) Trans-Pro / -65.3,-152.5	Allowed (0.6%) Cg_endo chi angles: 2.3,343,23.9	0.07Å	-	-	
A 3		THR	50.47	1.07Å HG23 with A 62 GLU CB	Allowed (0.66%) General / -169.1,117.8	Favored (2.6%) m chi angles: 316.8	0.31Å	OUTLIER(S) worst is CB--CG2: 4.0 σ	OUTLIER(S) worst is CA-C-O: 5.1 σ	
A 4		GLY	49.99	-	Favored (7.54%) Glycine / 130.4,-133.8	-	-	OUTLIER(S) worst is C--N: 12.1 σ	-	
A 5		THR	49.07	-	Favored (59.05%) General / -84.2,-4.8	OUTLIER (0.2%) chi angles: 208.8	0.06Å	-	-	
A 6		GLY	47.39	0.59Å O with A 59 THR O	Favored (3.98%) Glycine / -87.6,-133.3	-	-	-	-	
A 7		GLU	45.3	0.64Å HB3 with A 57 GLY HA2	Favored (10.86%) General / -133.9,112.0	OUTLIER (0.1%) chi angles: 42.7,223.2,333.8	0.08Å	-	-	
A 8		SER	42.1	0.72Å HB2 with A 61 GLU N	OUTLIER (0%) General / -59.4,-107.8	Allowed (1.6%) t chi angles: 202.1	0.12Å	-	-	
A 9		LYS	37.54	0.61Å O with A 10 CYS C	Favored (2.29%) General / -111.2,86.3	Allowed (1.5%) mppt chi angles: 296,87,59,186.3	0.05Å	-	OUTLIER(S) worst is N-CA-C: 7.0 σ	
A 10	A	CYS	30.04	0.61Å C with A 9 LYS O	OUTLIER (0.04%) Pre-Pro / -14.7,96.5	OUTLIER (0.1%) chi angles: 236.8	0.16Å	OUTLIER(S) worst is C--N: 5.1 σ	OUTLIER(S) worst is CA-C-N: 5.0 σ	
A 10	B	CYS	30.04	0.61Å C with A 9 LYS O	OUTLIER (0.04%) Pre-Pro / -14.7,96.5	Allowed (1%) m chi angles: 269.5	0.16Å	OUTLIER(S) worst is CB--SG: 5.9 σ	OUTLIER(S) worst is CA-CB-SG: 7.2 σ	
A 11		PRO	24.68	0.47Å HD3 with A 8 SER HB3	Favored (16.22%) Trans-Pro / -72.4,-27.3	Favored (72.1%) Cg_endo chi angles: 27.4,327.5,25	0.08Å	-	-	
A 12		LEU	18.63	-	Favored (12.48%) General / -140.8,119.5	Favored (60.8%) tp chi angles: 178,64.8	0.11Å	-	-	
A 13	A	MET	19.6	-	Favored (50.8%) General / -128.8,149.2	Favored (7.5%) tmm chi angles: 199.1,268.2,283.8	0.07Å	-	-	
A 13	B	MET	30.76	-	Favored (50.8%) General / -128.8,149.2	Favored (7.2%) ttp chi angles: 199.1,206.3,80.5	0.07Å	OUTLIER(S) worst is CG--SD: 7.9 σ	OUTLIER(S) worst is CG-SD-CE: 7.0 σ	
A 14	A	VAL	12.00	-	Favored (58.95%)	Favored (50.6%) t	0.11Å	-	-	

Bad Clashes >= 0.4 Angstrom:

B 2	PRO	HD3	B 50	SER	HB3	:1.096
A 3	THR	HG23	A 62	GLU	CB	:1.072
A 3	THR	HG23	A 62	GLU	HB2	:1.066
B 72	GLU	HG2	B 792	HOH	O	:0.763
B 2	PRO	HG2	B 48	LYS	O	:0.741
B 5	THR	HG21	B 44	PHE	O	:0.721
B 5	THR	HG22	B 45	ALA	HB2	:0.721
A 8	SER	HB2	A 61	GLU	N	:0.718
A 8	SER	HB2	A 61	GLU	H	:0.701
A 3	THR	CG2	A 62	GLU	HB2	:0.699
A 3	THR	HG23	A 62	GLU	HB3	:0.696
A 117	SER	HB2	B 117	ASER	HB2	:0.692
A 63	GLU	OE1	A 664	HOH	O	:0.684
A 119	BTHR	HG22	B 115	SER	OG	:0.673
B 72	GLU	CG	B 792	HOH	O	:0.661
A 99	ASP	OD1	A 724	HOH	O	:0.646

92.8% (232/250) of all residues were in favored (98%) regions.

96.8% (242/250) of all residues were in allowed (>99.8%) regions.

There were 8 outliers (phi, psi):

- A 2 Pro (-65.3, -152.5)
- A 8 Ser (-59.4, -107.8)
- A 10 Cys (-14.7, 96.5)
- A 125 Pro (-38.1, 179.0)
- A 126 Lys (135.8, 153.5)
- B 2 Pro (-101.3, -167.1)
- B 9 Lys (-66.7, 21.5)
- B 125 Pro (-54.5, -151.3)

Rama-Z (Ramachandran plot Z-score):

Interpretation: bad |Rama-Z| > 3; suspicious 2 < |Rama-Z| < 3; good |Rama-Z| < 2.

Scores for whole/helix/sheet/loop are scaled independently;

therefore, the values are not related in a simple manner.

whole: -0.54 (0.50), residues: 250

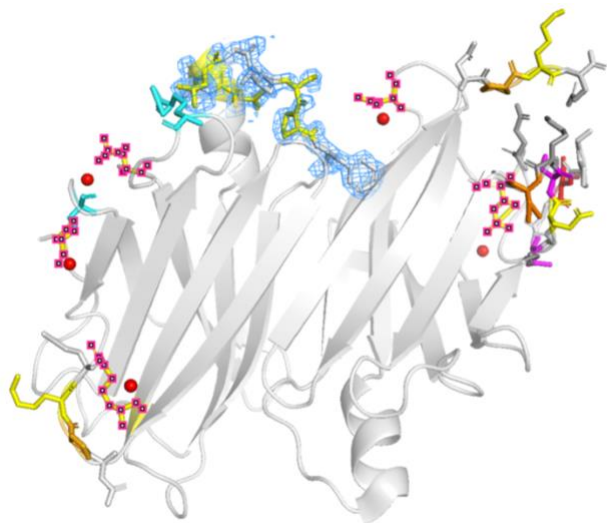
helix: 0.04 (0.94), residues: 14

sheet: 1.41 (0.47), residues: 108

loop: -2.16 (0.49), residues: 128

Part III: Refinement & Correction (PyMOL)

1. Successful corrections



A. N-terminal conflict on chain A

Thr3 had critical overlap with main helix, Thr5 was a rotamer outlier. Changing rotation of Thr3 allowed side group look toward solvent, eliminating 1.07Å conflict. Thr5 was fitted to 2Fo-Fc map (red and violet residues).

B. Loop 80-87 optimization on chain B

Cluster of errors included Lys80, Leu82, Ser85 and Pro86. Correction of some outliers (unfortunately, I don't remember which of them). There was some changes due to breaking the structure during rotations of residues (yellow sticks with blue density map).

C. **Artifact water molecules**, eg HOH792, created phantom conflicts with Glu72 and Asp74. Removal of 6 water molecules reduced water with clasher count from 20 to 14 (red balls).

2. Trial and error corrections

Important note that not all corrections were successful, some interventions leads to worse statistics, so changes were reversed. Eg, Pro2 and Ser50 1.096Å conflict by rotating rotamers. But clashscore rose to 16.03 after that, and number of broken bonds increased to 24 too. Its can be caused by rigid prolin ring which doesn't allow freely angles. So, changes was cancelled to keep structural integrity.

Summary statistics

All-Atom Contacts	Clashscore, all atoms:	16.03		45 th percentile* (N=1784, all resolutions)
	Clashscore is the number of serious steric overlaps (> 0.4 Å) per 1000 atoms.			
Protein Geometry	Poor rotamers	21	9.21%	Goal: <0.3%
	Favored rotamers	184	80.70%	Goal: >98%
	Ramachandran outliers	8	3.20%	Goal: <0.05%
	Ramachandran favored	232	92.80%	Goal: >98%
	Rama distribution Z-score	-0.54 ± 0.50		Goal: abs(Z score) < 2
	MolProbity score [^]	2.90		27 th percentile* (N=27675, 0Å - 99Å)
	Cβ deviations >0.25Å	1	0.43%	Goal: 0
	Bad bonds:	24 / 2028	1.18%	Goal: 0%
	Bad angles:	58 / 2763	2.10%	Goal: <0.1%
Peptide Omegas	Cis Prolines:	0 / 16	0.00%	Expected: ≤1 per chain, or ≤5%
Low-resolution Criteria	CaBLAM outliers	7	2.8%	Goal: <1.0%
	CA Geometry outliers	4	1.63%	Goal: <0.5%
Additional validations	Chiral volume outliers	0/310		
	Waters with clashes	20/186	10.75%	See UnDowser table for details

In the two column results. the left column gives the raw count. right column gives the percentage.

3. Results

As result, it was possible to achieve some improvement of 1TTA quality. Quantative indicators showed transition from low quality (31st percentile) to above average (53rd percentile).

Metric	Start value	Final value	Interpretation
Clashscore	16.29	14.22	▼ reduction of steric strain
Percentile Rank	31st	53rd	▲ structure is better than 53% of other structures with similar resolution
Poor Rotamers	22 (9.65%)	20 (8.77%)	▼ local geometry was corrected
Waters with clashes	20	14	▼ hydration level optimized, artifact water removed
MolProbity Score	2.92	2.83	▼ overall quality score improvement

Summary statistics

All-Atom Contacts	Clashscore, all atoms:	14.22		53 rd percentile* (N=1784, all resolutions)
	Clashscore is the number of serious steric overlaps (> 0.4 Å) per 1000 atoms.			
Protein Geometry	Poor rotamers	20	8.77%	Goal: <0.3%
	Favored rotamers	184	80.70%	Goal: >98%
	Ramachandran outliers	8	3.20%	Goal: <0.05%
	Ramachandran favored	232	92.80%	Goal: >98%
	Rama distribution Z-score	-0.54 ± 0.50		Goal: abs(Z score) < 2
	MolProbity score [^]	2.83		30 th percentile* (N=27675, 0Å - 99Å)
	Cβ deviations >0.25Å	1	0.43%	Goal: 0
	Bad bonds:	24 / 2028	1.18%	Goal: 0%
	Bad angles:	58 / 2763	2.10%	Goal: <0.1%
Peptide Omegas	Cis Prolines:	0 / 16	0.00%	Expected: ≤1 per chain, or ≤5%
Low-resolution Criteria	CaBLAM outliers	7	2.8%	Goal: <1.0%
	CA Geometry outliers	4	1.63%	Goal: <0.5%
Additional validations	Chiral volume outliers	0/310		
	Waters with clashes	14/180	7.78%	See UnDowser table for details

In the two column results, the left column gives the raw count, right column gives the percentage.

4. Limitations

Despite on some success during refining of 1TTA protein, there are some clear limitations that prevented to achieve 90th percentile:

- Backbone strain issue: Ramachandran outliers (8) and bad bonds (24) remains high, due to PyMol Mutagenesis wizard which can change only side chains (rotaners), but not main chain.
- Local optimum: correcting rotamers often led to domino effect – when corrected one, other conflict was created.
- Symmetry: due to tetramer structure of protein, errors in chain A mirrored to C, but here I considered only on chains A and B, and not duplicated changes to C and D chains.

Conclusions

Initial structure contained a number of steric conflicts, rotamer anomalies and water artefacts which typical for old models without careful verification.

Electron density visualization allowed to identify and fix critical false configurations (eg Thr3), with high clashscore.

Cleaning the structure from false-positive waters is an effective and quick method to improve statistics without affecting protein structure.

Obtained model with clashscore=14.22 is suitable for further usage, but needed to achieve perfect geometry (<5.0) – so, real-space refinement in Coot/Phenix needed to move main chain.